PATRIKIOS CONSTANTI

Massapequa, NY
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EDUCATION

Massapequa High School Ames Campus 2023-2024

Massapequa High School 2024–2027

Honors: LILT Spanish Poetry 2nd Place (2026); Seal of Biliteracy Candidate; AP Scholar with Distinction

RESEARCH EXPERIENCE

Independent Mars ISRU Research | 2024–Present

• Developed a theoretical model exploring CO-assisted reduction of SiO_2 as a potential supplement to MOXIE oxygen production

• Contacted 20+ researchers and engineers for technical feedback and feasibility critique

• Submitted manuscript for publication consideration and presented findings at local science events

CUNY STEM Research Academy 2026

• Designed and modeled a Martian rotorcraft using aerodynamic calculations and 3D CAD

• Presented research at the American Museum of Natural History symposium

• Continued independent development of the project after the program

LEADERSHIP

Founder | Massapequa SPARKS Chapter 2026–Present

• Founded a STEM outreach organization providing hands-on science activities and academic support for younger students
• Recruited and managed a 5-member board and coordinated student volunteers
• Organized STEM lessons, tutoring, and fundraising activities

National Honor Society Service Leader 2026–Present

• Elected service leader for a 283+ member National Honor Society Chapter
• Started Volunteer of the Month, helped introduce STEM related events, and contributed to monthly newsletter

EMPLOYMENT

Host/Cashier | Nautilus Diner 2022–Present

• Assist with front-of-house operations, customer service, and cash handling
• Provide Spanish-English workplace translation when needed; Tutor Spanish-speaking coworkers in academic subjects

ACTIVITIES

Martial Arts | 2022–2026
Japanese Jujitsu, Brazilian Jiu Jitsu, Kickboxing

Science Olympiad | 10th & 12th Grade
Participated in engineering events including Hovercraft and Egg Drop.

SKILLS

Technical: Autodesk Fusion, 3D Printing & Prototyping, Python, Microsoft Office, Google Workspace
Research: Technical Writing, Literature Review, Quantitative Modeling, Experimental Design
Training: NASA Open Science Essentials
Languages: Spanish, Greek
Communication: Public Speaking, Research Presentation